USC Dornsife
Dana and David Dornsife
College of Letters, Arts and Sciences
Spatial Sciences Institute

SSCI 214g (35635)
Human Populations and Natural Hazards

Syllabus

Units: 4

Term – Day – Time: Spring 2026; Tuesday and Thursday, 9:30-10:50am

Locations:
Lecture: SGM 101
Discussion Sections:

- Tuesday 11-11:50am @ DMC 203
- Wednesday 10-10:50am @ MHP 102
- Thursday 11-11:50am @ KAP 167
- Friday 10-10:50am @ WPH 101

Instructor: Darren Ruddell, PhD, GISP
Office: AHF B57F
Regular Office Hours: Mondays, 1:30-2:30pm and Wednesdays 12-1pm PT, and by appointment via email
Contact Info: druddell@usc.edu, 213-740-0521

Library Help: Andy Rutkowski
Office: LIPA B40-A
Office Hours: By appointment via email
Contact Info: arutkows@usc.edu

IT Help: Spatial Support
Contact Info: spatial_support@usc.edu

Course Scope and Purpose

Human populations and natural hazards (earthquakes, hurricanes, floods, drought) are increasingly in conflict throughout the world, as witnessed by the destructive forces of hurricanes on coastal cities and the effects of rainfall variability on agriculture in semi-arid lands. While natural hazards represent ongoing processes and functions of the Earth as a living organism, they can also be human-induced processes or events. Problems arise when people live in regions of the world where hazards exist with the potential to cause loss (i.e., life, injury, property, or other). As the global population surpasses eight billion inhabitants, increased competition for land and resources has driven people to live in more remote environments and higher density areas than ever before, resulting in increased human exposure and vulnerability to the risk of environmental hazards. While some individuals or communities are resilient, meaning they possess a high capacity to absorb impacts and recover from a hazardous event, others, such as poor residents living in inner-city slums or rural dwellers located on marginal land, struggle to cope, recover, or rebuild from a disaster.

This course examines the complex and coupled relationship between human development (population growth, urbanization) and environmental hazards by exploring a range of topics, such as: What do hazard, risk, vulnerability, and disaster mean, and how are these terms measured? What do hazards have to do with human values? How is exposure to environmental hazards different in developing and industrialized countries? And what responsibility does the government have to protect individuals from risk? Students will utilize quantitative and qualitative methods – including geospatial technologies – to gain insight into these questions – where and why hazards occur – and the subsequent impacts disaster events have on the social world (such as mortality, displacement, property damage, or other losses). Students will reflect on how society evaluates and confronts the dangers posed by natural hazards, and how political, economic, and/or cultural settings can serve to attenuate or exacerbate human vulnerability before, during, or after a disaster occurs.

SSCI 214g fulfills the requirements of the USC General Education program, Social Analysis (Category C) and is designed to serve students of diverse backgrounds and academic interests (e.g., anthropology, earth sciences, environmental studies, human health, international relations, public policy, and spatial sciences, among others). It is appropriate for students with or without an extensive background in science.

Learning Outcomes

Students who excel in SSCI 214g will be able to:

- Discuss human populations and societies in the context of geographic location and exposure to the earth's natural hazards.
- Describe the underlying processes that give rise to natural hazards such as earthquakes, volcanic eruptions, hurricanes, landslides, and how these affect human populations.
- Explain how society evaluates and confronts the dangers posed by natural hazards from political, social, and ethical perspectives.

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- Utilize geospatial technologies to visualize the locations and dangers of natural hazards to human populations.
- Discuss the consequences and outcomes of environmental hazards for human populations.
- Compare and contrast technological innovations used to monitor, predict, and warn society about natural hazards and impending disasters.

Students may vary in their competency levels on these abilities. You can expect to acquire these abilities only if you honor all course policies, attend classes regularly, complete all assigned work in good faith and on time, and meet all other course expectations of you as a student.

Prerequisite(s): None
Co-Requisite(s): None
Concurrent Enrollment: None
Recommended Preparation: None

Class Conduct

Harassment, sexual misconduct, interpersonal violence, and stalking are not tolerated by the university. All faculty and most staff are considered Responsible Employees by the university and must forward all information they receive about these types of situations to the Civil Rights Coordinator. The Civil Rights Coordinator is responsible for assisting students with supportive accommodations, including academic accommodations, as well as investigating these incidents if the reporting student wants an investigation. The Office of Civil Rigths Compliance (OCRC) is also responsible for coordinating supportive measures for transgender and nonbinary students such as faculty notifications, and more. If you need supportive accommodations you may contact the Civil Rights Coordinator directly (ocrc@usc.edu or 213-740-5086) without sharing any personal information with me. If you would like to speak with a confidential counselor, Relationship and Sexual Violence Prevention Services (RSVP) provides 24/7 confidential support for students (213-740-9355 (WELL); press 0 after hours).

USC Culture Journey is an initiative aimed at fostering a more **inclusive, values-based, and integrity-driven culture** across the university. The components of the Culture Journey include: community engagement, core values, strategic cultural goals, ongoing assessment, and leadership commitment. Use the following link to see full details on the Culture Journey: https://culturejourney.usc.edu/

Course Structure

This course aims to engage students in the dynamic processes underlying earth systems and natural hazards as well as the impacts that disaster events present to human populations. Student learning experiences are achieved through a combination of lectures, assignments, discussions, and course readings. Lectures, complemented by readings, will present core concepts, provide detailed explanations of assignments, and include activities such as individual and collaborative problem definition and problem-solving experiences. Discussions will complement lectures with opportunities to interactively explore lecture topics in greater depth.

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For example, in discussions students will be encouraged to ask questions and participate in group conversations that will allow a broader and deeper understanding of natural hazards and disasters to emerge. No make-up dates will be offered for missed quizzes or exams, so mark the appropriate dates on your calendars. If there is a legitimate conflict, speak with a course instructor as soon as possible so we can make alternative arrangements.

Technological and Communication Requirements

This course will use ArcGIS Online (AGOL) to investigate human populations and natural hazards whereby students will locate and explore various spatial datasets that offer unique and innovative insights in hazards research. The modeling software and geospatial data required for course assignments will be accessed using computing resources provided by the Spatial Sciences Institute. Every student must a computer with a fast Internet connection and an up-to-date web browser.

Brightspace – This course will use the Brightspace learning management system, which allows students to access course content, upload assignments, participate in discussion forums, among other learning experiences.

Communications – All materials to be handed in will be submitted via Brightspace. It is each student's responsibility to stay informed about what is going on in the course. In addition to email about time-sensitive topics, all announcements will be posted on the News page in Brightspace. Students should read all email sent from Brightspace or from course instructor as soon as possible. Please double-check to be sure that mail sent from both Brightspace and the instructor does not go into your junk mail. The instructor will endeavor to respond to all email within 24 hours of receipt, aiming for no more than 72 hours delay. In the rare case that an instructor is off-line for an extended period, an announcement will be posted to the class Brightspace site.

Required Readings and Supplementary Materials

The required reading materials for this course are:

- Smith, K. 2024. *Environmental Hazards: Assessing Risk and Reducing Disaster*, 7th Edition. NY, NY: Routledge.
- Flannery, T. 2005. *The Weather Makers.* NY, NY: Grove Press.
- Hardin, G. 1968. The Tragedy of the Commons. *Science* Volume 162, No. 3859, 1243-1248.

Description and Assessment of Assignments

Your grade in this class will be determined on the basis of several different assessments:

In-Class Activities – 5 worth a total of 5 points. Students will complete five in-class activities to promote engagement with course content and active participation and discussion during class lecture meetings.

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Homework Assignments – 6 worth a total of 30 points. Students will be required to complete six homework assignments comprised of quantitative and/or qualitative analysis to gain insight on the physical processes underlying natural hazards and disasters as well as examine the impact these events have on human populations.

Hazard and Disaster Log – 1 worth a total of 10 points. Students will be required to keep a journal of five significant natural hazard and/or disaster events that happened over the course of the semester. Working in pairs, students will report the Hazard & Disaster Log in the form of a digital Story Map (available through AGOL) and will present in class at the end of the semester. This assignment will require students to locate and evaluate technical information from online agency sites such as the U.S. Geological Survey (USGS), the National Oceanic and Atmospheric Administration (NOAA), among others.

Discussions – 10 worth a total of 20 points. Structured weekly discussion sections are highly interactive and will focus on combinations of theory and practice to promote deeper learning of core concepts. Students are required to participate in discussion meetings on weekly content. Each student must register for a separate discussion section.

Midterm Exam – 1 worth a total of 15 points. The midterm exam consists of multiple choice, short answer, and simple problem questions. Students are expected to take the exam at the indicated time.

Final Exam – 1 worth a total of 20 points. The final exam will consist of multiple choice, short answer, simple problem questions, and a short essay. Students are expected to take the exam at the indicated time.

Grading Breakdown

Assessment	Number	Points Each	Total Points
In-Class Activities	5	1	5
Homework Assignments	6	5	30
Hazard & Disaster Log	1	10	10
Discussion Sections	10	2	20
Midterm Exam	1	15	15
Final Exam	1	20	20
Total	**24**		**100 points**

Grading Scale

Assignments in this and other SSCI courses, are graded on the letter grade scale where A is exemplary, B is very good, C is satisfactory, D is unsatisfactory, and F needs improvement. Final grades use the same letter grade scale with C being the minimum passing grade for credit at the graduate level. The grading scale follows:

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A	> 93 points		B-	80-82 points		D+	67-69 points
A-	90-92 points		C+	77-79 points		D	63-66 points
B+	87-89 points		C	73-76 points		D-	60-62 points
B	83-86 points		C-	70-72 points		F	<60 points

Assignment Submission Policy

Assignments will be submitted for grading via Brightspace using the due dates specified in the Course Schedule below.

Students are expected to attend and participate in all lecture and lab sessions and to complete and upload all assignments by the deadline detailed in the Course Schedule. Late work will be assessed a penalty of 10% per day and zero grades will be assigned for work that is more than seven days late.

SSI Policy on the Creation of Original Work and Use of Generative AI

All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. Students may not have another person or entity complete any substantive portion of an assignment or reuse work prepare for courses without obtaining written permission from the instructor(s). Developing strong competencies in research, writing, and the technical execution of geospatial technologies are foundational to SSI academic programs that are designed to prepare you for success in the workplace. Therefore, using generative AI tools – unless explicitly specified otherwise – is strictly prohibited in this course, will be identified as plagiarism, and will be reported to the Office of Academic Integrity.

Grading Timeline

My goal is to provide grading and feedback on each course assignment in time for you to take my feedback into consideration as the course progresses. Generally, this means that you can expect feedback within 1 week after a given assignment's due date.

Learning Experience Evaluations

Please note Learning Experience Evaluations for the course take place at the end of the semester and are facilitated by the University. These evaluations provide an important review of student experiences in the course.

Course Content Distribution and Synchronous Session Recordings Policies

USC has policies that prohibit recording and distribution of any synchronous and asynchronous course content outside of the learning environment.

Recording a university class without the express permission of the instructor and announcement to the class, or unless conducted pursuant to an Office of Accessibility Services (OSAS) accommodation. Recording can inhibit free discussion in the future, and thus infringe on

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the academic freedom of other students as well as the instructor. (Living our Unifying Values: The USC Student Handbook, page 13).

Distribution or use of notes, recordings, exams, or other intellectual property, based on university classes or lectures without the express permission of the instructor for purposed other than individual or group study is prohibited. This includes but is not limited to providing materials for distribution by services publishing course materials. This restriction on unauthorized use also applies to all information, which has been distributed to students or in any way has been displayed for use in relationship to the class, whether obtained in class, via email, on the internet, or via any other media. (Living our Unifying Values: The USC Student Handbook, page 13).

Schedule

	Topic	Readings and Assignments	Deliverables and Due Dates
Module 1: The Nature of Human Populations			
Week 1 1/13 and 1/15	**Introduction to Human Populations, Natural Hazards and Disasters** Lectures: Introduction to human populations and the associated impacts of natural hazards; Human life, tracking past and present migration patterns, and the evolution of environmental hazard paradigm perspectives. No discussion meeting in Week 1.	Reading: Smith (2024) Ch. 1	No deliverables
Week 2 1/20 and 1/22	**Human Dimensions and Spatial Analysis** Lectures: Spatial patterns in the context of impacts on human populations; human-environment interactions and anthropogenic factors to hazards. Discussion 1	Reading: Smith (2024) Ch. 2; Flannery (2005) Ch. 1-4 Homework Assignment 1	No deliverables
Week 3 1/27 and 1/29	**Vulnerability, Resilience, and Sustainability** Lectures: Complexity science and sustainability science; Global links between physical and socio economic systems and recording loss; introduction to GIS for measuring human vulnerability and economic loss. Discussion 2	Reading: Smith (2024) Ch. 3; Flannery (2005) Ch. 5-8 Homework Assignment 2	Homework Assignment 1 due 1/28 by 11:59pm

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	Topic	Readings and Assignments	Deliverables and Due Dates
Week 4 2/3 add 2/5	**Geoaccounting and the Commons** Lectures: Geoaccounting ecosystem services to measure and model resources; the commons and ecological health. Discussion 3	Reading: Hardin (1968); Flannery (2005) Ch. 9-12 Homework Assignment 3	Homework Assignment 2 due 2/4 by 11:59pm
Week 5 2/10 and 2/12	**Risk Assessment and Management** Lectures: Assessing and mitigating risk with geospatial data and GIS infrastructures; human risk perception. Discussion 4	Reading: Smith (2024) Ch. 4; Flannery (2005) Ch. 13-16 Hazard & Disaster Log	Homework Assignment 3 due 2/11 by 11:59pm
Week 6 2/17 and 2/19	**Disaster Risk Reduction** Lectures: Mitigation and adaptation strategies to reduce the impacts of disasters induced by climate change. Discussion 5	Reading: Smith (2024) Ch. 5; Flannery (2005) Ch. 17-24	No deliverables
Week 7 2/24 and 2/26	**Midterm Exam** Prepare for and complete Midterm Exam. No discussion in Week 7		Complete Midterm Exam in class on 2/26
Module 2: Natural Hazards and Impacts on Humans			
Week 8 3/3 and 3/5	**Tectonic Hazards: Earthquakes, Tsunamis, Volcanoes and Mass Movement** Lectures: The processes that produce tectonic hazards – earthquakes, tsunamis, volcanoes, and mass movements – and how these events impact society. Discussion 6	Reading: Smith (2024) Ch. 6, 7, 8 & 9; Homework Assignment 4	No deliverables
Week 9 3/10 and 3/12	**Flood and Drought Hazards** Lectures: Tropical cyclones – formation, classification, human populations at risk, and mitigating resulting storm damage; introduction	Reading: Smith (2024) Ch. 11 & 12; Flannery (2005) Ch. 25-30	Homework Assignment 4 due 3/11 by 11:59pm

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	Topic	Readings and Assignments	Deliverables and Due Dates
	to extreme weather events and related outcomes. Drivers and impacts of drought. Discussion 7	Homework Assignment 5	
	Spring Recess 3/16-3/20		
Week 10 3/24 and 3/26	**Wildfires and Extreme Temperature Hazards** Lectures: Introduction to wildfires and the Wildland Urban Interface (WUI) and the feedback loop between human development and wildfire risk. Introduction to extreme temperatures and trends. Discussion 8	Reading: Smith (2024) Ch. 13 & 14; Flannery (2005) Ch. 31-35 Homework Assignment 6	Homework Assignment 5 due 3/25 by 11:59pm
Week 11 3/31 and 4/2	**Disease Epidemics and Human Rights** Lectures: An introduction on disease epidemics and human rights through the lens of spatial data Discussion 9	Reading: Smith (2024) Ch. 15	Homework Assignment 6 due 4/1 by 11:59pm
Module 3: Environmental Hazard and Risk in the Anthropocene			
Week 12 4/7 and 4/9	**Anthropocene Challenges and Technological Hazards** Lectures: "Man-made accidents", the third industrial revolution, case studies showing societal impacts, and options for the future. Discussion Optional – prepare Hazard and Disaster Log	Reading: Smith (2024) Ch. 16	No deliverables
Week 13 4/14 and 4/16	**Industrial and Waste Hazards** Lectures: The third industrial revolution and the impact on the environment. Discussion 10a: Present Hazard and Disaster Log.	Reading: Smith (2024) Ch. 17	Hazard & Disaster Log Presentation in Discussion Sections
Week 14 4/21 and 4/23	**The Future of our World** Lecture: Case studies showing societal impacts and options for the future. Discussion 10b: Present Hazard and Disaster Log.	Reading: Smith (2024) Ch. 18	Hazard & Disaster Log Presentation in Discussion Sections

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	Topic	Readings and Assignments	Deliverables and Due Dates
Week 15 4/28 and 4/30 Friday, 5/1 is the last day of class	**Using Hazards Knowledge for Action** Lectures: Class closure and Final Exam review. Discussion 10c: Present Hazard and Disaster Log.	Reading: Smith (2024) Ch. 19	Hazard & Disaster Log Presentation in Discussion Sections
Final Exams 5/6-5/13	**Final Exam** Complete Final Exam at scheduled time		Final Exam scheduled for 5/12 from 8-10am

Statement on Academic Conduct and Support Systems

Academic Integrity

The University of Southern California is a learning community committed to developing successful scholars and researchers dedicated to the pursuit of knowledge and the dissemination of ideas. Academic misconduct, which includes any act of dishonesty in the production or submission of academic work, comprises the integrity of the person who commits the act and can impugn the perceived integrity of the entire university community. It stands in opposition to the university's mission to research, educate, and contribute productively to our community and the world.

All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. You may not submit work written by others (including AI generated) or "recycle" work prepared for other courses without obtaining written permission from the instructor(s).

Other violations of academic integrity include, but are not limited to, cheating, plagiarism, fabrication (e.g., falsifying data), collusion, knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage.

The impact of academic dishonesty is far-reaching and is considered a serious offense against the university. All incidences of academic misconduct will be reported to the Office of Academic Integrity and could result in outcomes such as failure on the assignment, failure in the course, suspension, or even expulsion from the university.

For more information about academic integrity see the student handbook or the Office of Academic Integrity's website, and university policies on Research and Scholarship Misconduct.

Please ask your instructor if you are unsure what constitutes unauthorized assistance on an exam or assignment, or what information requires citation and/or attribution.

Students and Disability Accommodations:

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USC welcomes students with disabilities into all of the University's educational programs. The Office of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

Support Systems:

Counseling and Mental Health *(213) 740-9355 – 24/7 on call*

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

988 Suicide and Crisis Lifeline - 988 for both calls and text messages – 24/7 on call

The 988 Suicide and Crisis Lifeline (formerly known as the National Suicide Prevention Lifeline) provides free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week, across the United States. The Lifeline is comprised of a national network of over 200 local crisis centers, combining custom local care and resources with national standards and best practices. The new, shorter phone number makes it easier for people to remember and access mental health crisis services (though the previous 1 (800) 273-8255 number will continue to function indefinitely) and represents a continued commitment to those in crisis.

CARE-SC - (213) 740-9355(WELL) – 24/7 on call

Free and confidential therapy services, workshops, and training for situations related to gender- and power-based harm (including sexual assault, intimate partner violence, and stalking).

Office for Civil Rights Compliance - (213) 740-5086

Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

The Office of Student Accessibility Services (OSAS) - (213) 740-0776

OSAS ensures equal access for students with disabilities through providing academic accommodations and auxiliary aids in accordance with federal laws and university policy.

USC Campus Support and Intervention - (213) 740-0411

Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

USC Culture Journey - (213) 740-8258

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At USC our six unifying values support our University's mission, and guide our behaviors, the way we engage with ourselves and our community, to create our Trojan culture.

USC Emergency - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call

Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety - UPC: (213) 740-6000, HSC: (323) 442-1200 – 24/7 on call

Non-emergency assistance or information.

Office of the Ombuds - (213) 821-9556 (UPC) / (323-442-0382 (HSC)

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

Occupational Therapy Faculty Practice - (323) 442-2850 or otfp@med.usc.edu

Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.

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